

PRACTICAL 5:

Practical5(1)

Problem Phase:

Manage a playlist using a **doubly linked list** with operations: add at beginning, add at end, insert after a song, remove first song, count, and display.

Solution:

- Each node contains song, prev, and next.
- For insertion after a song, find the song and update both next and prev links.
- Extra DLL step:
temp->next->prev = newNode;
- If the song is not found → print "**Song not found**" and do not insert.
- Removing the first song requires updating head and its prev pointer.

Practical5(2)

Problem Phase:

The problem is to manage students sitting in a **circle** using both a **singly circular linked list** and a **doubly circular linked list**. Students can join, leave, and the current circle can be displayed. After every operation, the circle must remain connected.

Solution:

In a **singly circular linked list**, when a student leaves, the previous student's next is connected to the next student. In a **doubly circular linked list**, both next and prev links are updated. This keeps the circle unbroken.

